

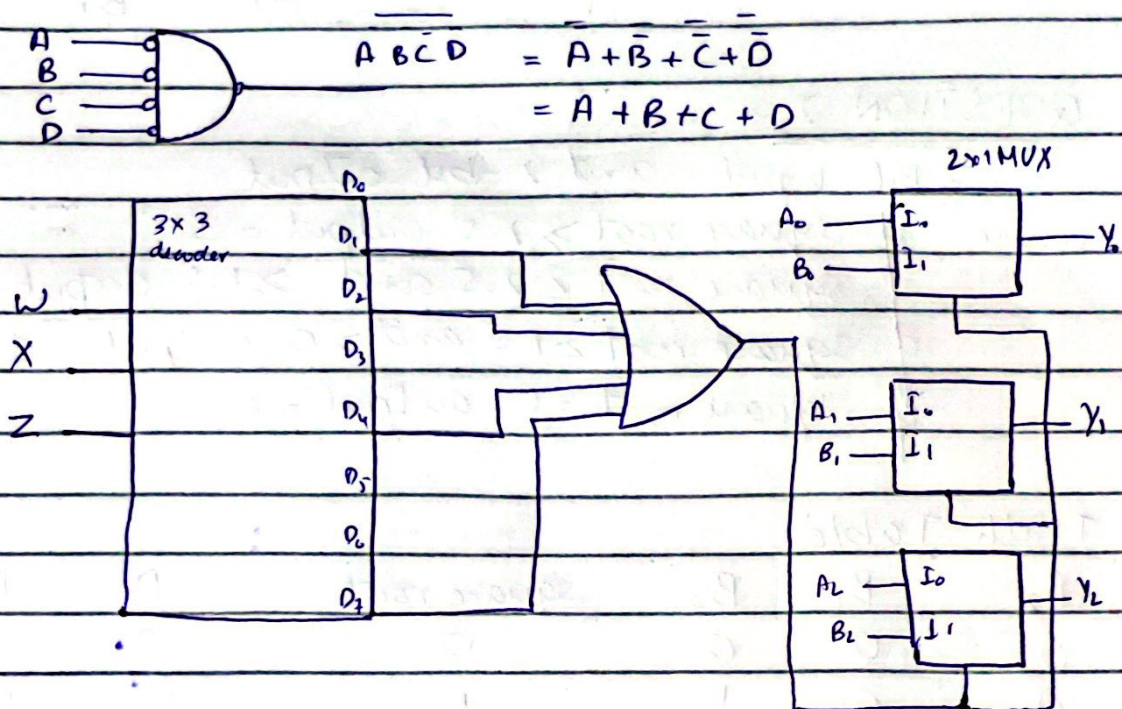
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ROLL NO. :- 22L-7651

SECTION :- 4A (BSEE)

ASSIGNMENT :- 2

QUESTION 1:-



Truth table:

W	X	Z	S	Y_2	Y_1	Y_0
0	0	0	0	A_2	A_1	A_0

0	0	Q	Q	B ₂	B ₁	B ₀
0	1	0	1	B ₂	B ₁	B ₀
0	1	1	0	→ A ₂	A ₁	A ₀
1	0	0	1	B ₂	B ₁	B ₀
1	0	1	0	→ A ₂	A ₁	A ₀
1	1	0	0	A ₂	A ₁	A ₀
1	1	1	1	B ₂	B ₁	B ₀

QUESTION 2 :-

3-bit input and 2-bit output

where if square root ≥ 2.5 output = 3

if square root ≤ 2.5 and ≥ 1.5 output = 2

if square root < 1.5 and > 0 output = 1

if square root = 0 output = 0

Truth Table:

B ₂	B ₁	B ₀	square root	D ₁	D ₀
0	0	0	0	0	0
0	0	1	1	0	1
0	1	0	1.414	0	1
0	1	1	1.732	1	0
1	0	0	2	1	0
1	0	1	2.236	1	0
1	1	0	2.449	1	0
1	1	1	2.646	1	1

Now we should design D₁ & D₀ using 4x1 MUX and few additional gates

If we observe

$B_2, B_1 = 0, 0$ output $D_1 = 0$

$B_2, B_1 = 0, 1$ output $D_1 = B_0$

$B_2, B_1 = 1, 0$ output $D_1 = 1$

$B_2, B_1 = 1, 1$ output $D_1 = 1$

similarly for output D_0 .

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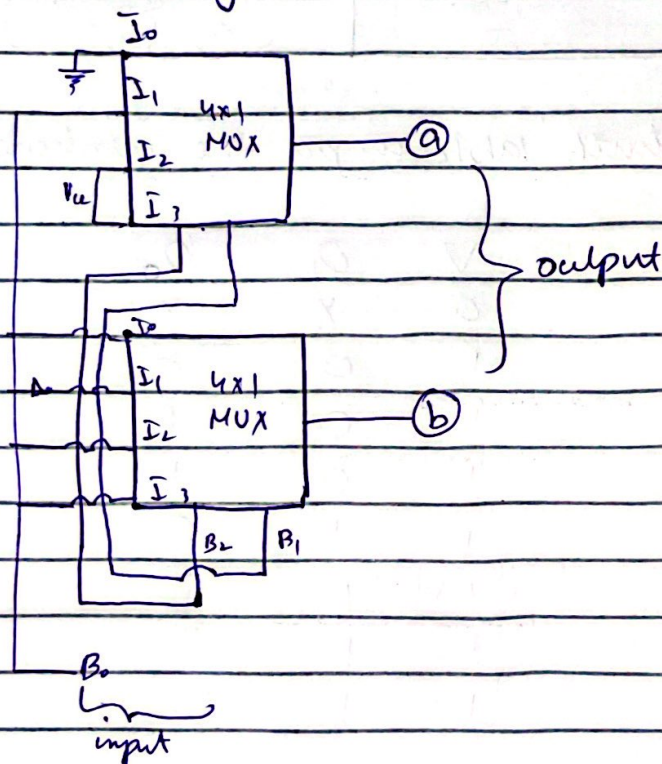
$B_2, B_1 = 0, 0$ output $D_0 = B_0$

$B_2, B_1 = 0, 1$ output $D_0 = \overline{B_0}$

$B_2, B_1 = 1, 0$ output $D_0 = 0$

$B_2, B_1 = 1, 1$ output $D_0 = B_0$

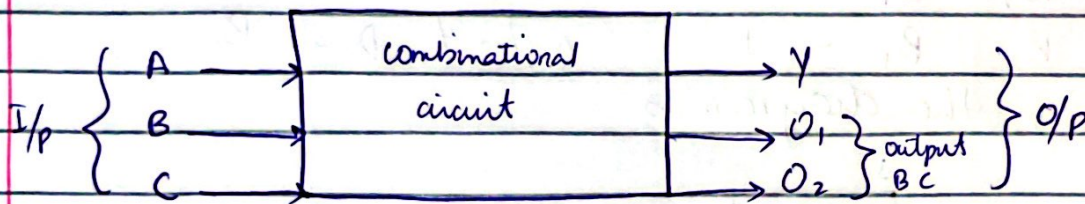
So the diagram is



QUESTION 3:-

- let us consider that the A, B, C are the input variables to represent the 3-bit pattern
- O_1 and O_0 are the output variables which represent the 2-bit binary code
- And V is an additional output as per mentioned in the problem.

Hence the block diagram of the combinational circuit can be drawn as



Construct the truth table as per the mentioned specification as:

A	B	C	V	O_1	O_0
0	0	0	0	X	X
0	0	1	1	0	0
0	1	0	1	0	1
0	1	1	1	0	1
1	0	0	1	1	0
1	0	1	1	1	0
1	1	0	1	1	0
1	1	1	1	1	0

Implementation of combinational circuit using dual 4x1 MUX,

→ The SOP expression of the output variable can be obtained as

$$V = \sum m(1, 2, 3, 4, 5, 6, 7)$$

$$O_0 = \sum m(2, 3) + d(0)$$

and

$$O_1 = \sum m(4, 5, 6, 7) + d(0)$$

It is observed that output O_1 is equal to input A .

$$O_1 = A$$

Implementation of output 'V' for 4x1 MUX, :-

	I_0	I_1	I_2	I_3
$\bar{A}$	0	①	②	③
A	④	⑤	⑥	⑦
A	1	1	1	1

Hence

$$I_0 = A$$

$$I_1 = 1$$

$$I_2 = 1$$

$$I_3 = 1$$

Implementation of output 'O₀' for 4x1 MUX, :-

	I_0	I_1	I_2	I_3
$\bar{A}$	0	1	②	③
A	4	5	6	7
	0	0	$\bar{A}$	$\bar{A}$

Hence

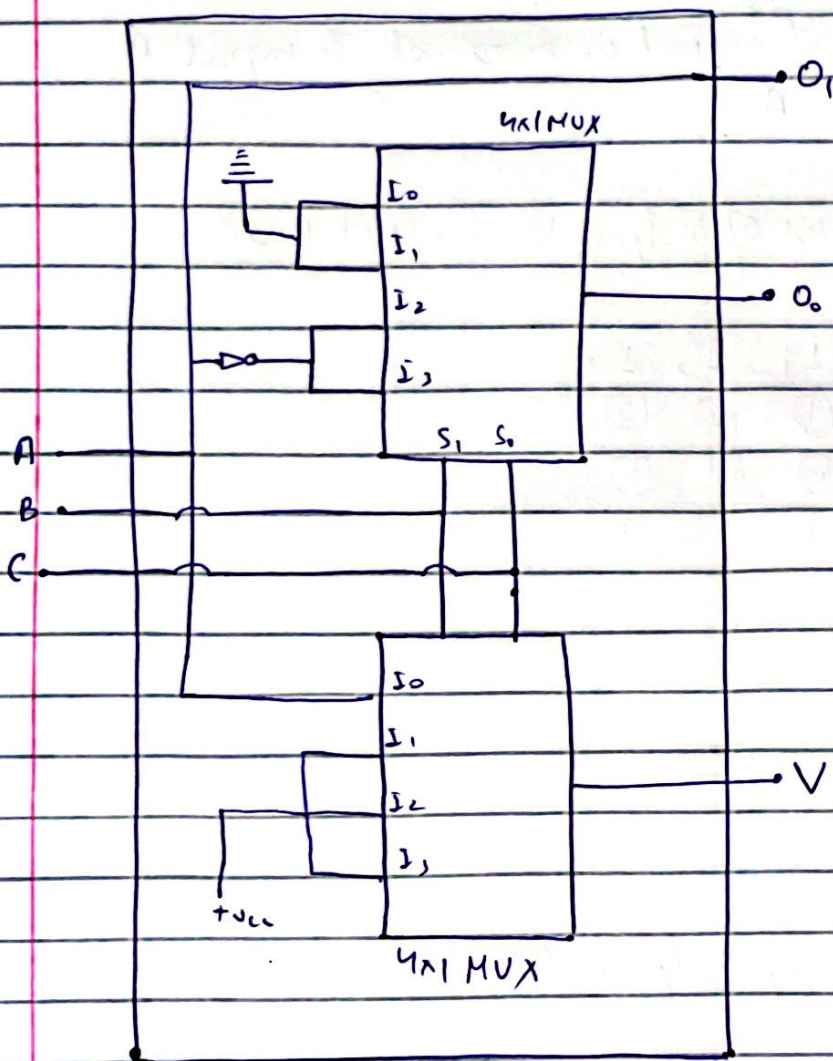
$$I_0 = 0$$

$$I_1 = 0$$

$$I_2 = \bar{A}$$

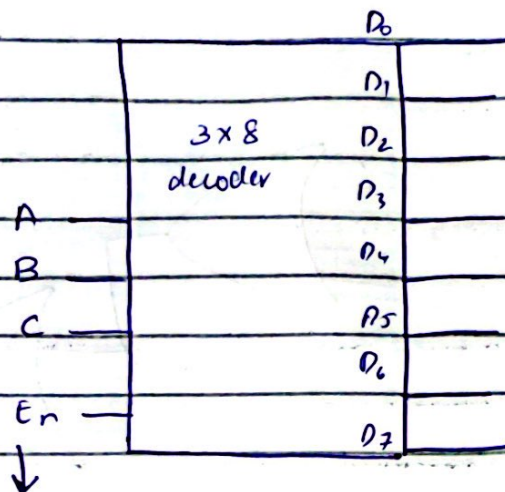
$$I_3 = \bar{A}$$

Combinational Circuit:-

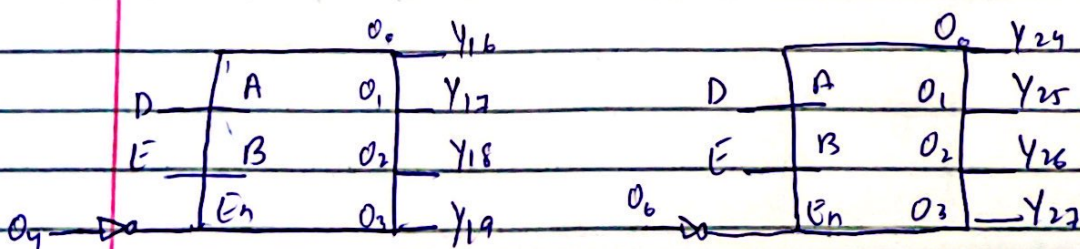
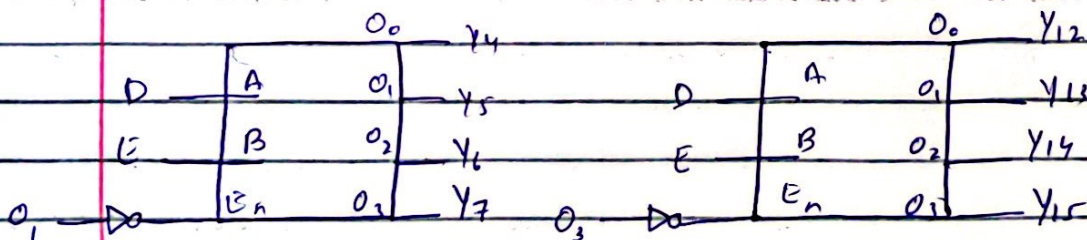
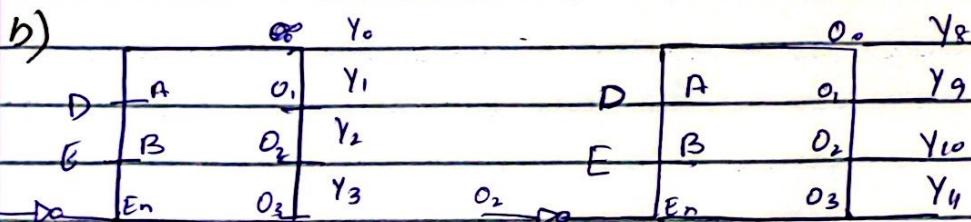


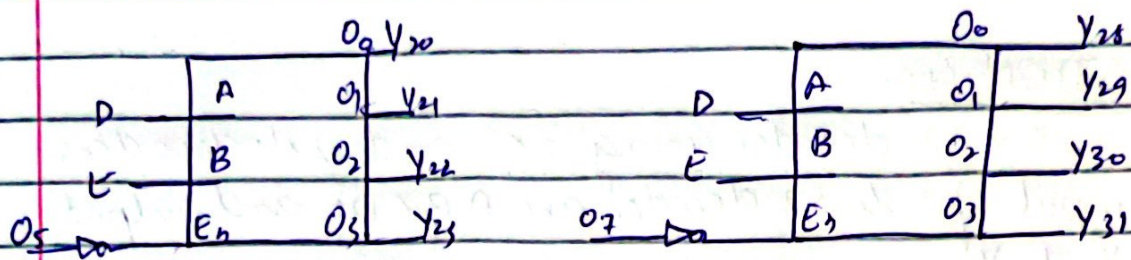
QUESTION 4:-

a) 5×32 decoder using 3×8 and 2×4 decoders
 Let input of 5 to 32 decoder are $A B C D E$ and output are Y_0 to Y_{31} .



this is the global active enable signal for 3×8 decoder





c)

